How-To Library

Informatica

Profiling Configuration Recommendations in Enterprise Data Catalog

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Abstract

When you enable data discovery for a resource and scan the resource, Enterprise Data Catalog identifies profiling-related metadata, such as null values, distinct values, inferred data types, unique keys, and data domains in the resource. This article provides information on supported connections, authentication methods, troubleshooting, and performance metrics to run data discovery on resources.

Supported Versions

- Enterprise Data Catalog 10.5 x

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Supported Connections

Before you enable data discovery for a resource in Catalog Administrator, you need to create a connection to the resource in Informatica Administrator. You can create a native, JDBC, or ODBC connection to the resource.

The following tables list the connections that support data discovery for each resource type:

Cloud

Resource	Native	JDBC	ODBC
Amazon Redshift	Supported	Supported, Not Certified	Supported, Not Certified
Amazon Redshift Spectrum	Supported	Supported, Not Certified	Not supported
Amazon S3	Supported	Not supported	Not supported
Athena	Not supported	Supported	Not supported
Apache Cassandra	Supported	Supported, Not Certified	Supported, Not Certified
Azure Microsoft SQL Data Warehouse	Supported	Not supported	Supported, Not Certified
Azure Microsoft SQL Server	Use MSSQL Server connection to run data discovery.	Not supported	Supported, Not Certified
Google Big Query	Supported	Not supported	Not supported
Microsoft Azure Data Lake Store Gen1	Supported	Not supported	Not supported
Microsoft Azure Data Lake Store Gen2	Supported	Not supported	Not supported
Microsoft Azure Blob Storage	Supported	Not supported	Not supported
Snowflake	Supported	Supported, Not Certified	Supported, Not Certified
Vertica	Not supported	Supported	Not supported
Salesforce	Supported	Not supported	Not supported

Data Engineering

Resource	Native	JDBC	ODBC
HDFS	Supported	Not supported	Not supported
Hive	Supported	Not applicable	Not applicable

File Management

Resource	Native
Local File System	Supported
File system using SFTP	Supported
File system using SMB/CIFS	Supported

Resource	Native
OneDrive	Supported
SharePoint	Supported

Database Management

Resource	Native	JDBC	ODBC
IBM DB2	Supported	Supported, Not Certified	Supported, Not Certified
IBM DB2 for z/OS	Supported	Not supported	Not supported
IBM Netezza	Supported, Not Certified	Supported, Not Certified	Supported
Microsoft SQL Server	Supported	Supported, Not Certified	Supported, Not Certified
Oracle	Supported	Supported	Supported, Not Certified
SAP HANA	Not supported	Not supported	Supported
SAP S/4 HANA	Not supported	Not supported	Supported
Sybase	Not supported	Not supported	Supported
Teradata	Supported, Not Certified	Supported, Not Certified	Supported

Application

Resource	Native	JDBC	ODBC
SAP R/3	Supported	Not applicable	Not applicable
MySQL	Not applicable	Supported	Not applicable

Supported Authentication Methods

The following table lists the sources and supported authentication methods to connect to the source:

Resource	Authentication Method	Data Discovery Support
Amazon S3	Basic Authentication	Supported
	IAM Authentication	Supported but not certified
	Temporary security credentials through Assume Role	Supported but not certified
Microsoft Azure Blob Storage	Shared Key	Supported

Resource	Authentication Method	Data Discovery Support
	Shared Access Signature	Supported
Microsoft Azure Data Lake Store Gen1	Shared Access Key	Supported
Microsoft Azure Data Lake Store Gen2	Shared Access Key Active Directory from version 10.4.1.1 onwards	Supported

Supported Sampling Options

When you run a profile on a source, you can choose to run the profile on all the rows or on a subset of rows in the resource.

You can choose one of the following sampling options to determine the number of rows to run the profile job on:

- All rows. Runs the profile on all the rows in the data source.
- Auto random rows. Runs the profile on a random sample of rows.
- Random N rows. Runs the profile on the configured number of random rows.
- First N rows. Runs the profile on the first N number of rows in the resource.
- Limit N rows. Runs the profile based on the number of rows in the data object. This option is available in Spark runtime environment.
- Random Percentage. Runs the profile on a percentage of rows in the data object. This option is available in Spark runtime environment.

The following tables list the resources and supported sampling options to run the profile:

Cloud

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
Amazon Redshift	Supported. Supports SQL pushdown to database from version 10.4.1.1 onwards.	Supported. Supports SQL pushdown to database from version 10.4.1.1 onwards. Not recommended to use in versions prior to 10.4.1.1.	Not supported	Supported but not recommended	Not applicable	Not applicable
Amazon Redshift Spectrum	Supported	Supported but not recommended	Not applicable	Not applicable	Not applicable	Not applicable

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
Amazon S3	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
Apache Cassandra	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
Athena	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
Azure Microsoft SQL Data Warehouse	Supported. Supports SQL pushdown to database from version 10.4.1.1 onwards.	Supported but not recommended	Not supported	Supported but not recommended	Not applicable	Not applicable
Azure Microsoft SQL Server	Supported	Supported	Not supported	Supported but not recommended	Not applicable	Not applicable
Google Big Query	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
Microsoft Azure Data Lake Store Gen1	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
Microsoft Azure Data Lake Store Gen2	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
Microsoft Azure Blob Storage	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
Snowflake	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
Salesforce	Supported	Supported but not recommended.	Not supported	Supported but not recommended.	Not applicable	Not applicable
Vertica	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable

Data Engineering

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
HDFS	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
Hive	Supported	Not applicable	Supported, Recommended	Not applicable	Not applicable	Not applicable

File Management

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
Local File System	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
File System using SFTP	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
File System using SMB/ CIFS	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
OneDrive	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable
SharePoint	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Supported only for CSV files.	Not applicable	Not applicable

Database Management

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
IBM DB2	Supported	Supported	Not supported	Supported	Not applicable	Not applicable
IBM DB2 for z/OS	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
IBM Netezza	Supported	Supported	Not supported	Supported	Not applicable	Not applicable
Microsoft SQL Server	Supported Supports SQL pushdown to database.	Supported	Not supported	Supported	Not applicable	Not applicable
Oracle	Supported	Supported	Supported only on the Spark engine.	Supported Not supported on the Spark engine.	Supported	Not applicable

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
SAP Hana	Supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not supported	Supported but not recommended. The profiling service fetches all rows for the profile run.	Not applicable	Not applicable
Sybase	Supported	Supported	Not supported	Supported	Not applicable	Not applicable
Teradata	Supported Supports SQL pushdown to database.	Supported Supports SQL pushdown to database.	Not supported	Supported	Not applicable	Not applicable

Application

Resource	First N rows	Random N rows	Random Percentage	Auto Random rows	Limit N rows	All rows
SAP R/3	Supported	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
MySQL	Supported	Supported	Not supported	Supported	Not applicable	Not applicable

Supported Runtime Environments

You can choose to run a profile in the native runtime environment, on the Blaze engine, or Spark engine. When you choose the Blaze engine or Spark engine, select a Hadoop connection to run the profiles.

Note: You must create a separate cluster to run the Blaze engine, or Spark engine. You cannot run them on the Informatica Hadoop Service.

The following tables list the resources and recommended and supported runtime environments to run the profile:

Cloud

Resource	Native	Blaze Engine	Spark Engine
Amazon Redshift	Recommended	Supported, Not Certified	Not applicable
Amazon Redshift Spectrum	Supported	Supported, Not Certified	Supported
Amazon S3	Recommended	Supported	Certified only for partitioned and non-partitioned Parquet files.

Resource	Native	Blaze Engine	Spark Engine
Apache Cassandra	Recommended	Supported, Not Certified	Not applicable
Athena	Recommended	Supported, Not Certified	Not applicable
Azure Microsoft SQL Data Warehouse	Recommended	Supported, Not Certified	Not applicable
Azure Microsoft SQL Server	Recommended	Supported, Not Certified	Not applicable
Google Big Query	Recommended	Supported, Not Certified	Not applicable
Microsoft Azure Data Lake Store Gen1	Recommended	Supported, Not Certified	Not applicable
Microsoft Azure Data Lake Store Gen2	Recommended	Supported, Not Certified	Certified only for partitioned and non-partitioned Parquet files.
Microsoft Azure Blob Storage	Recommended	Supported, Not Certified	Not applicable
Snowflake	Recommended	Supported, Not Certified	Not applicable
Salesforce	Recommended	Supported, Not Certified	Not applicable
Vertica	Recommended	Supported, Not Certified	Not applicable

Data Engineering

Resource	Native	Blaze Engine	Spark Engine
HDFS	Supported	Recommended	Supported for Avro, XML, JSON, and partitioned Parquet files on CDH and CDP clusters. Certified for CSV files. Only CSV files are processed through Spark engine. Other files are processed through Blaze engine or native runtime environment for profile run.
Hive	Supported	Supported (if Spark runtime environment is not available)	Recommended

File Management

Resource	Native	Blaze Engine	Spark Engine
Local File System	Recommended	Supported, Not Certified	Not applicable
File System using SFTP	Recommended	Supported, Not Certified	Not applicable
File System using SMB/CIFS	Recommended	Supported, Not Certified	Not applicable

Resource	Native	Blaze Engine	Spark Engine
OneDrive	Recommended	Supported, Not Certified	Not applicable
SharePoint	Recommended	Supported, Not Certified	Not applicable

Database Management

Resource	Native	Blaze Engine	Spark Engine
IBM DB2	Recommended	Supported	Not applicable
IBM DB2 for z/OS	Recommended	Not supported	Not applicable
IBM Netezza	Recommended	Supported	Not applicable
Microsoft SQL Server	Recommended	Supported	Not applicable
Oracle	Recommended	Supported	Supported and certified for Sqoop connection.
SAP HANA	Recommended	Supported, Not Certified	Not applicable
Sybase	Recommended	Supported	Not applicable
Teradata	Recommended	Supported	Not applicable

Application

Resource	Native	Blaze Engine	Spark Engine
SAP R/3	Recommended	Supported, Not Certified	Not applicable
MySQL	Recommended	Supported	Not applicable

Performance Metrics

The performance of Enterprise Data Catalog depends on the size of data that needs to be processed. The data set size, sampling options, number of data domains to infer, and runtime environment determines the performance of data discovery for resources.

Note: The profiling execution time and throughput vary based on the following factors:

- Location of the data source. The data source can be on the cloud or on a local file system.
- Data source type
- Volume of content in the tables or files
- Number of tables, rows, and columns
- Column length
- File type
- Average and maximum file size

- Number of unique and duplicate datasets
- Sampling mode
- Number of data domains
- Source system hardware configuration
- Disk latency
- Network latency between source systems

The following tables list the tested time taken to run a column profile with data domain discovery for ten data domains on different runtime environments, and the hardware and parallelism details. The time taken to run profile depends on the resource type, its data set size, and the sampling option:

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
Amazon S3	600,000 partitioned parquet files Total data set size of 25 GB.			No	Spark engine: 6 hours		7 node CDH cluster. Core: 32 RAM: 128 GB	7 node CDH cluster. Core: 32 RAM: 128 GB
	100,000 files which includes unstructured and semi-structured files.		All rows	No	Native engine: 24 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Apache Cassandra	1000 tables with 20 columns each		First N: 1000	No	Native engine: 32 minutes	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
	10 tables with an average of 20 million records		First N: 10,000,000	No	Native engine: 1 hour	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
	1 table with 90 million rows		All rows	No	Native engine: 3 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
	1000 tables with 100 columns each	Column Profile and Data Domain Discovery Validated with Personally Identifiable Information (PII) and Personal Health Information (PHI) data domain groups.	First N: 10,000 rows	No	Native engine: 1 hour	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Google Big Query	10,000 tables 100 columns/ table and 10,000 rows/ table	Column Profile and Data Domain Discovery Validated with 10 data domains.	First N: 10,000 rows	Yes	Native engine: 6.5 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
	1 table with 25 columns and 33 million rows	Column Profile and Data Domain Discovery Validated with 10 data domains.	All rows	Yes	Native engine: 5.5 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
	1 table with 25 columns and 33 million rows	Column Profile and Data Domain Discovery Validated with 10 data domains.	First N: 100,000 rows	Yes	Native engine: 10 minutes	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
HDFS	10,000 CSV files	Column Profile and Data Domain Discovery Validated with 10 data domains.	First N: 10,000 rows	No	Native engine: 2 hours Blaze engine: 2.5 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB
Hive	1 table with 195 columns and 4 billion rows		All rows	No	Spark engine: 8 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB
	Random percentage: 10%			No	Blaze engine: 5 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB
	200 tables 81 columns/ table, 100 million/ table		All rows	No	Blaze engine: 36 hours Spark engine: 22 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
	Random percentage : 10%			No	Blaze engine: 8 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB
	10,000 tables 100 columns/ table, 10,000 rows/ table		All rows	No	Blaze engine: 3 hours Spark engine: 2.5 hours		3 node CDH cluster. Each node: Core: 32 RAM: 128 GB	3 node CDH cluster. Each node: Core: 32 RAM: 128 GB
Microsoft SQL Server	10,000 tables 100 columns/ table		First N: 10,000 rows	No	Native engine: 2 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Oracle	90,000 tables with 100 columns each		First N: 10,000 rows	No	Native engine: 12 hours Blaze engine: 3 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz	3 node CDH cluster: Each node: Core: 32 RAM: 128 GB DIS configuration: Core: 32 RAM: 128 GB	3 node CDH cluster: Each node: Core: 32 RAM: 128 GB DIS configuration: Core: 32 RAM: 128 GB

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
	1 table with 100 million rows		All rows	No	Native engine: 1.5 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Snowflake	1000 tables 195 columns/ table, 1 million rows/ table		First N: 1,000,000 rows All rows	No	Native engine: 15 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
SAP R3	63,000 tables with an average of 20 columns/ table		All rows	No	Native engine: 40 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Teradata	1000 tables 100 columns/ table		First N: 10,000 rows	No	Native engine: 4 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
File system on SFTP	10,000 mixed files	Data Domain Discovery Validated with 10 data domains.	All rows	No	Native engine: 30 mins	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		

Resource	Data Set	Profile Run Option	Sampling Option	Run Similarity Profile	Time Taken in Runtime Environment	Hardware and Parallelism		
						Native Engine	Blaze Engine	Spark Engine
Amazon Redshift	1000 tables	Column Profile	First N: 10,000 rows	No	Native engine: 30 mins	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
Azure Data Lake Storage Gen2	10,000 mixed files	Column Profile and Data Domain Discovery Validated with Personally Identifiable Information (PII) data domain group.	All rows	No	Native engine: 2 hours	Core: 32 RAM:128 GB Parallelism:20 Processor speed: 3.2 GHz		
	100,000 Part files in 15,000 folders	Column Profile and Data Domain Discovery Validated with 6 data domains.	All rows	No	Databricks engine: 20 mins			

Note: For better performance, use First N as the Sampling Option and limit the number of rows to 10,000 for all resources.

Incremental profiling

Incremental profiling is available for all resource types except the relational database sources.

Performance Tuning Parameters for Profiling

Tuning profiling performance involves configuring the Data Integration Service parameters, Data Integration Service concurrency parameters, and profiling warehouse parameters.

Tuning Data Integration Service Parameters for Profiling

To optimize data profiling, you can configure Data Integration Service parameters, Data Integration Service profiling tuning parameters, and Data Integration Service grid configuration. You can also tune unique key inference performance parameters.

Note: For optimal performance, configure Data Integration Service on a standalone node or multiple nodes with horizontal scaling instead of a grid.

You need to configure the Temporary Directories and Maximum Execution Pool Size parameters for the Data Integration Service. You can configure parameters, such as Reserved Profile Threads that apply to the Profiling Service Module. Before you use the parameter recommendations, verify that you have identified a node or grid and the requirement is to configure the node or grid optimally to run profiles.

The following configuration parameters have an impact on different components of the profiling and discovery installation. You can increase or decrease the values for these parameters based on the performance requirements:

Parameter	Description
Maximum Execution Pool Size	The maximum number of requests that the Data Integration Service can run concurrently. Requests include data previews, mappings, and profiling jobs. This parameter has an impact on the Data Integration Service.
Maximum Profile Execution Pool Size	The total number threads to run profiles. This parameter has an impact on the Profiling Service Module.
Temporary Directories	Location of temporary directories for the Data Integration Service process on the node. This parameter has an impact on the Data Integration Service machine.

Note: For more information about Data Integration Service parameters, see the *Profiling and Discovery Sizing Guidelines* article.

Data Integration Service Profiling Tuning Parameters

The following table lists the values for the Data Integration Service profiling tuning parameters:

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
Maximum Concurrent Columns	100	5	Maximum number of columns merged in single mapping.
Maximum Column Heap Size	512	512	Maximum cache size allocated for a mapping.
AdvancedProfilingServiceOptions.ColumnsThreshold	100	Default	Maximum number of columns merged for a mapping. Applicable only for column profiling.

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
AdvancedProfilingServiceOptions.DomainMappingColumnThreshold	100	50	Maximum number of columns merged for a mapping. Applicable only for data domain discovery.
AdvancedProfilingServiceOptions.MaxDataDomains	150	Default	Maximum number of data domains combined for a data domain discovery mapping.
ExecutionContextOptions.Optimizer.MaxMappingPorts	-1	Default	Controls platform optimization during mapping execution.
ExecutionContextOptions.Optimizer.PushdownType	none	Default	Controls pushdown optimization during mapping execution.
ExecutionContextOptions.Optimizer.ExtractSourceConstraints	false	Default	Controls extracting constraint details for a relational source during mapping execution.
ExecutionContextOptions.JVMOption1	1024M	1024M	Controls mapping process heap size.
ExecutionOptions.MaxProcessLifeTime	1800000	Default	Controls mapping process life time.
ExecutionContextOptions.Optimizer.NativeExpressionTx	false	Default	Controls SISR optimization during mapping execution.

Data Integration Service Node Scaling for Profiling

The execution time and throughput for data discovery varies based on the following parameters. It also depends on the hardware configuration of the Data Integration Service node.

- Volume of data in the tables and files
- Number of rows
- Number of unique or duplicate data sets
- Number of columns
- Column length
- Sampling options
- Number of data domains

You can choose a configuration from the following options to scale the Data Integration Service node for profiling:

- **Option 1 — Vertical hardware scaling with a standalone node**
  - This option is recommended if you have a single profiling resource.

- The throughput for a single profiling resource can be increased by vertically scaling the node hardware configuration.
- To achieve vertical scaling, you must calculate the Data Integration Service concurrency parameters for profiling and increase the number of CPU cores and system memory on the node. For information on how to calculate the Data Integration Service concurrency parameters, refer to the "Data Integration Service Concurrency Parameters" on page 21.

- **Option 2 — Horizontal hardware scaling with multiple nodes**
  - This option is recommended if you have multiple profiling resources.
  - The throughput for multiple profiling resources can be increased by horizontally scaling the node hardware configuration. This configuration is used to increase the throughput across resources.
  - Create multiple nodes with dedicated profiling warehouse schema that can be selected at the resource level under the domain connection settings.
- **Option 3 — Grid configuration**
  - For optimal performance, configure Data Integration Service on a standalone node or multiple nodes with horizontal scaling instead of a grid.
  - This option is used to increase the throughput for a single profiling resource by horizontal scaling.
  - Using this configuration, you can improve the data discovery throughput by up to 1.4x by increasing the total hardware capacity by 2x across multiple grid nodes.
  - The generation of mappings and compilation is performed on the master node. The execution takes place on both the master and the secondary nodes. Therefore, the hardware utilization on the master node is always higher than the secondary nodes.
  - The minimum hardware requirement recommended for a node used in a grid configuration is 32 CPU cores and 64 GB memory.
  - For information on how to configure multiple nodes in a grid configuration, refer to the "Data Integration Service Grid Configuration for Enterprise Data Catalog" on page 20 section.

Data Integration Service Grid Configuration for Enterprise Data Catalog

In Enterprise Data Catalog, to use the Data Integration Service on a grid, you need to configure a shared directory with read and write access on all the nodes in the grid.

To configure the *<INFA_HOME>/<profile_temp_folder>* shared directory on all the nodes, perform the following steps:

1. Log in to Informatica Administrator.
2. Navigate to **Manage** > **Services and Nodes** > **Domain Navigator**.
3. Select the grid in the **Domain Navigator**.
4. In the **Properties** > **Custom Properties** section, click the **Edit** icon.
5. In the **Edit Custom Properties** dialog box, click **New**.
6. In the **New Custom Property** dialog box, enter the following details:
   a. In the **Name** field, enter **AdvancedProfilingServiceOptions.CustomFlag8**.
   b. In the **Value** field, enter the temporary folder. For example, `/data2/profile_temp`.
7. Recycle the Data Integration Service.

In Enterprise Data Catalog, you can use the shared directory to perform the following tasks:

- To stage XML or JSON schemas on a Data Integration Service node
- To stage unstructured files and results on a Data Integration Service node
- To stage similarity results on a Data Integration Service node

Tuning Performance for Unique Key Inference Process

The unique key inference throughput and execution time are exponential and they depend on the following factors:

- Schema size
- Data set volume
- Profiling sample size
- Profiling concurrency

To tune unique key inference performance, configure the **ExecutionContextOptions.JVMoption1 = -Xmx4096M** JVM option in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Data Integration Service Concurrency Parameters

To increase the profiling performance throughput, you can tune the following Data Integration Service concurrency parameters:

- Maximum On-Demand Execution Pool Size
- Maximum Profile Execution Pool Size
- Maximum Concurrent Profile Jobs
- Maximum Concurrent Profile Threads
- AdvancedProfilingServiceOptions.FileConnectionLimit. This parameter applies to SAP, XML, and JSON sources.
- AdvancedProfilingServiceOptions.HiveConnectionLimit. This parameter applies to Hive sources.
- Maximum # of Connections for source connection

You can configure the concurrency parameters in one of the following ways:

- In the Administrator tool, navigate to the Data Integration Service properties to configure the concurrency parameters.
- In the command prompt, navigate to the `<INFA_HOME>/isp/bin/plugins/ps/concurrencyParameters/` location.
  The *ConcurrencyParameters.properties* file in this location contains the default values of the Data Integration Service concurrency parameters for each load type. Run the *setConcurrencyParameters* script with the load type option to update the parameter values simultaneously for Data Integration Service.

Calculate Concurrency Parameter Values

You can calculate the value for concurrency parameters based on the number of cores. It is recommended that you multiply the number of cores with 0.6 and round off the value to the nearest whole value. For example, if the number of cores configured for the Data Integration Service is 16, then the concurrency parameter value is calculated as follows:16*0.6=9.6. The concurrency parameter value is rounded off to the nearest whole value, namely, 10.

The following table describes how to calculate the Data Integration Service concurrency parameters for profiling based on the number of cores:

Concurrency Parameter	Value*	Description
Maximum On-Demand Execution Pool Size	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service that includes all types of mappings.
Maximum Profile Execution Pool Size	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service.
Maximum Concurrent Profile Jobs	Number of cores x 0.6	Maximum parallel tables or files processed.
Maximum Concurrent Profile Threads	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service for File sources.
AdvancedProfilingServiceOptions.FileConnectionLimit (For CSV, XML, and JSON metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for File sources.
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for Hive sources.
Maximum # of Connections	Number of cores x 0.6	Total number of database connections for each resource.
* Indicates the value of the parameter if the number of rows is less than or greater than 100, 000 rows with sampling of rows enabled or disabled.		

setConcurrencyParameters Script

You can run the script to update all the parameter values in the *ConcurrencyParameters.properties* file simultaneously. After you run the script, the script stops the Data Integration Service, updates the parameter values in the Data Integration Service, and then enables the Data Integration Service.

The *Maximum # Connection Pool Size* parameter is configured at the data source level. Therefore, you need to update this parameter in the Administrator tool for relational sources.

The setConcurrencyParameters script uses the following syntax:

```
setConcurrencyParameters.sh
[small/medium/large]
[DomainName]
[DISName]
[Username]
[Password]
```

The following table describes the setConcurrencyParameters script arguments:

Argument	Description
small/medium/large	Required. Enter small, medium, or large.
DomainName	Required. Enter the domain name.

Argument	Description
DISName	Required. Enter the Data Integration Service name.
Username	Required. Enter the user name that the Data Integration Service uses to access the Model Repository Service.
Password	Required. Enter the password for the Model repository user.

Usage

```
./setConcurrencyParameters.sh [small/medium/large] [DomainName] [DISName] [Username]
[Password]
```

Example

```
./setConcurrencyParameters.sh small domain DIS Administrator Administrator
```

Default Parameter Values for Small, Medium, and Large Options in the Script

Parameter	Small	Medium	Large
Data Integration Service server hardware	CPU cores is16. Memory is 32 GB.	CPU cores is 32. Memory is 64 GB.	CPU cores is 64. Memory is 128 GB.
Maximum On-Demand Execution Pool Size	10	20	40
Maximum Profile Execution Pool Size	10	20	40
Maximum Concurrent Profile Jobs	10	20	40
Maximum Concurrent Profile Threads	10	20	40
AdvancedProfilingServiceOptions.FileConnectionLimit (For SAP, XML, JSON sources)	10	20	40
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive sources)	10	20	40
Maximum # of Connections for source connection	10	20	40

Tuning Profiling Warehouse Parameters

The profiling warehouse stores profiling results. More than one Profiling Service Module may point to the same profiling warehouse. The main resource for the profiling warehouse is disk space.

The following table lists the parameters that you need to tune to improve the performance of the profiling warehouse based on the size of data:

Parameter	Low	Medium	High	Description
Tablespace	50 GB	200 GB	500 GB	The maximum amount of disk space required to temporarily store the profiling results before these results get stored in the Enterprise Data Catalog.
CPU	4	8	16	The number of cores required by the profiling warehouse

Note: For more information about the profiling warehouse guidelines, see the Profiling and Discovery Sizing Guidelines article.

Tuning Performance for Profiling on Spark Engine

To increase the profiling throughput for high volume sources, run the profiles on the Spark engine in the Hadoop environment.

To optimize profiling performance, run the following resources on the Spark engine:

- HDFS sources
- Hive sources
- Partitioned parquet files on Amazon S3 and Microsoft Azure Data Lake Store Gen2.

To tune the profiling throughput, configure the following parameters:

- Increase the value of **spark.yarn.executor.memoryOverhead** from 384 to 2048.
- Configure the **spark.dynamicAllocation.maxExecutors** parameter to 5 for every 3 nodes. For example, you can configure **spark.dynamicAllocation.maxExecutors** to 10 for 6 nodes.

To configure the parameters, navigate to the **Informatica Administrator** > **Manage** > **Connections** > **Hadoop CCO connection** > **Spark Configuration** > **Advanced Properties** section.

The minimum system requirement for each node in the Spark cluster is 32 CPU cores and 128 GB of memory. The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling:

Load Type	Number of Cluster Nodes	Hive Schema Size	Number of Tables per Schema	Number of Columns per Table	Recommended Sampling	Number of Data Domains	SLA
Medium	3	Upto 10	100	100	1 million	10	48 - 72 Hours
High	6	Upto 100	200	100	10 million	10	1 - 4 Weeks

Load Type	Number of Cluster Nodes	Hive Schema Size	Number of Tables per Schema	Number of Columns per Table	Recommended Sampling	Number of Data Domains	SLA
Very High	12	Upto 100	1000	100	10 million	10	4 - 10 Weeks

Note:
- The SLA is based on Hive table execution performance on an on-premise HDP/CDH cluster.
- The recommended mode for profiling is random percentage sampling. The maximum recommended sampling for the Spark engine is 10 million. You can extrapolate the sampling based on the actual row count.
- The profiling execution time and throughput vary based on the following factors:
  - Location of the data source. The data source can be on the cloud or on a local file system.
  - Data source type
  - Volume of content in the tables or files
  - Number of tables, rows, and columns
  - Column length
  - File type
  - Average and maximum file size
  - Number of unique and duplicate datasets
  - Sampling mode
  - Number of data domains
  - Source system hardware configuration
  - Disk latency
  - Network latency between source systems
- The data domain discovery execution time is linearly proportional to the total number of data domains selected for profiling.
- The cluster performance will vary based on the underlying hardware configuration of the cluster nodes and the cluster type such as EMR/HDP/CDP.

For information about best practices and to tune the Spark engine, see the Tuning and Sizing Guidelines for Informatica® Data Engineering Integration 10.4.0 and Performance Tuning for the Spark Engine articles.

Tuning Performance for Profiling on Blaze Engine

To increase the profiling throughput for high volume sources, you can run the profiles on the Blaze engine in the Hadoop environment for relational sources that exceed 100 million rows.

The following table lists the custom properties for the Data Integration Service and the values you need to configure to run the profiles on the Blaze engine:

Parameter	Value
ExecutionContextOptions.Blaze.AllowSortedPartitionMergeTx	false
ExecutionContextOptions.GridExecutor.EnableMapSideAgg	true
ExecutionContextOptions.RAPartitioning.AutoPartitionResult	true
AdvancedProfilingServiceOptions.BlazeBatchSize	1
AdvancedProfilingServiceOptions.DomainMappingColumnThreshold	1
AdvancedProfilingServiceOptions.ColumnsThreshold	1

Note: When you run profiles on the Blaze engine, the profiling service stages the source data in the cluster nodes which uses higher disk space than the source size.

For information about best practices and to tune the Blaze engine, see the *Performance Tuning and Sizing Guidelines for Informatica Big Data Management* and *Best Practices for Big Data Management Blaze Engine* articles.

Memory Sizing to Run Profiles on Large Unstructured Files

You can run a profile on unstructured files, such as PDF, Microsoft Excel, and Microsoft Word files.

Memory Requirements to Run Profiles on Large PDF Data Sources

Large unstructured files in PDF format require more memory during profiling.

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the PDF file size:

PDF File Size	Memory (in MB)
Less than or equal to 25 MB	1024M
Less than or equal to 50 MB	2048M
Less than or equal to 100 MB	5120M

Memory Requirements to Run Profiles on Large Microsoft Excel or Microsoft Word Files

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the Microsoft Excel or Microsoft Word file size:

File Size	Memory (in MB)
Less than or equal to 5 MB	1024M
Less than or equal to 25 MB	2048M
Less than or equal to 100 MB	7168M

To tune the *ExecutionContextOptions.JVMOption1* parameter, navigate to the **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section in Informatica Administrator.

Tuning Resource Configuration Parameters for Profiling

Snowflake Resource

To optimize the profiling performance for Snowflake high volume resources, configure the following custom properties in Informatica Administrator:

- **AdvancedProfilingServiceOptions.ColumnsThreshold**. Set the value to **10**.
- **ExecutionContextOptions.JVMoption1**. Set the value to **-Xmx2048M**.

To configure these parameters, navigate to the **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Microsoft Azure Data Lake Store Gen2 Resource

You can configure the following parameters to optimize the profiling performance for Microsoft Azure Data Lake Store Gen2 resources:

Heap Size for pmdtmsvc2 Process

To set the maximum heap size for pmdtmsvc2 process used for profiling Microsoft Azure Data Lake Store Gen2 high volume resources in the native runtime environment, set the following custom property :

`ExecutionContextOptions.JVMoption1` custom property to -Xmx4096M value.

The pmdtmsvc2 process memory requirement might increase for a large file. When the configured heap size does not meet the file processing memory requirement, a heap size error appears and the mapping is marked as failed.

To configure the custom property, navigate to the **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Concurrent Threads

You can set concurrent threads to download the files from Microsoft Azure Data Lake Store Gen2 source. By default, this option is set to 10. It is recommended to configure this value in the range of current number of cores to twice the number of CPU cores. For example, on a 16 CPU cores server, you can configure this parameter from 16 through 32.

You can configure the following option at resource level to override the default value:

`-Dconnector.access.concurrent.thread=<value>`

To configure this JVM option at the resource level in Catalog Administrator, open the resource and set the custom option on the **Metadata Load Settings** tab.

Block Size

You can set the block size to download the files in chunks using multiple threads. By default, the option is set to 8MB.

You can configure this option at resource level to override the default value:

`-Dconnector.access.block.sizeInMB=<value>`

It is recommended that you consider the pmdtmsvc2 process heap size and concurrent threads to configure this option to an optimum value to avoid redundant network calls.

To configure this JVM option at the resource level in Catalog Administrator, open the resource and set the custom option on the **Metadata Load Settings** tab.

Heap Size

Configure the Data Integration Service heap size to 10GB in Informatica Administrator before you run the data domain discovery profiles in Catalog Administrator. You can increase this value if the data domains exceed 100.

Cassandra Resource

You can configure the following parameters to optimize the profiling performance for Cassandra resources:

Heap Size for pmdtmsvc2 Process

Configure the **ExecutionContextOptions.JVMoption1** custom property to -Xmx2048M to optimize the profiling performance for Cassandra high volume resources. This parameter sets the maximum heap size for pmdtmsvc2 process used for profiling.

To configure these parameters, navigate to **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Properties** > **Custom Properties** section.

Heap Size

Configure the Data Integration Service heap size to 10 GB when you run data domain discovery on Cassandra resources. Increase the heap size if the number of data domains exceed 100.

Profiling large tables on JDBC connections

Profiling large tables in MySQL, PostgreSQL, Vertica, Athena, and Databricks Delta using JDBC connectivity might cause performance issues as sampling is not supported on the native execution engine.

To resolve the issue, you can limit the number of rows fetched by the queries. Configure the following option at the resource level to specify the custom data profiling sampling value to 1000 and limit the number of rows to 1000: –`DcustomSampling='limit 1000'`.

To configure this option at the resource level in Catalog Administrator, open the resource and set the option on the **Metadata Load Settings** tab in the **Custom Options** property.

Profiling Recommendations for a Databricks Cluster

Verify the following recommendations to optimally run profiling on a Databricks cluster:

- The Databricks cluster and source data ,must reside in the same geographical region to reduce network latency.
- For better throughput and performance, you can enable autoscaling.

Hardware Recommendations for Databricks Runtime 5.5 LTS

Verify the following hardware recommendations:

- You can configure the worker or driver type node as Standard_DS3_v3 that includes 128.0 GB of memory, 32 CPU cores, and 6 DBU.
- You can configure a minimum of 1 worker node and a maximum of 8 worker nodes with autoscaling enabled.

Frequently Asked Questions

This section lists the frequently asked questions to solve issues related to profiles and data domain discovery.

Data Discovery

After I run the filesystem resource to perform data domain discovery on the source column that has Chinese characters in the column name, the catalog displays "?" instead of the column name. How do I resolve this issue?

To solve this issue, you can add the environment variable to the allowlist on the Cloudera manager page. For information about configuring environment variables on the Cloudera cluster, see the KB link: https://kb.informatica.com/solution/23/Pages/71/587165.aspx

After I create a Azure Data Lake Store Gen2 resource and run a profile on a source database that has UTF-16 characters, the profile run fails? How do I solve this issue?

You cannot run profile on the column that has UTF-16 characters. To solve this issue, change the UTF-16 characters to UTF-8 characters in the source database.

After I create the Cassandra resource and run a profile on the Cassandra database that contains large number of columns, the profile run fails. What can I do to resolve this issue?

The profile run may fail due to the large number of tombstones present in the Cassandra database. Before you run profile on Cassandra database, perform the following steps:

1. Go to the following location: `<cassandra_install_folder>/config/cassandra.yaml`
2. In the configuration setting of the `cassandra.yaml` file, set the `tombstone_failure_threshold` value as 10 million or more.

Does Informatica Data Quality resource support logical data sources for data discovery?

No. Informatica Data Quality does not support logical data sources.

When and where is the need to copy JDBC jars to externalJdbcJars directory

If you want to connect to sources through JDBC, you need to install and configure a JDBC Type 4 driver from a third-party vendor. To run profiles, copy the JDBC driver jar file to the following location:

`<Informatica installation directory>\externaljdbcjars`.

When do I use a JDBC connector for a resource?

You can use a JDBC connector if one of the following conditions are true:

- No native connector is available and you have a JDBC 4.x compliant driver.
- If the native connector has a limitation because of which you cannot you use it in your environment.

The following table shows a few use cases where you can use different connectors for Amazon Redshift resource:

Use Case	Connectivity Option
You do not want to use Amazon S3 as the staging directory because of incurring costs and extra privileges to access the staging area.	Use JDBC connection to connect to the resource.
You want to use incremental profiling.	Use native connection because JDBC does not support this function.
You want to infer data types.	Use native connection because JDBC infers boolean data type as an integer data type.
You want to improve the data discovery performance.	Use JDBC or native connections. You can experiment with either of the options to see which performs better in your environment.

Data Discovery on Spark engine

When I run a profile on the Spark engine, the profile run fails. How do I resolve this issue?

Before you run a profile, perform the following steps:

1. Create a JDBC warehouse connection.
2. Obtain the Data Direct JAR files of the database that you use to extract data.
3. Copy the files to the following JDBC folder :`<INFA_HOME>/externaljdbcjars`

When I perform data domain discovery on the Spark engine, it takes a long time to complete. How do I resolve this issue?

When you perform the data domain discovery on the Spark engine, the profiling service launches the Sqoop connection and imports the reference data from the Content Management Service to the Hadoop cluster. The data domain discovery occurs in multiple successive sessions and each session closes automatically.

Sometimes, the sessions take a long time to close because of the OJDBC driver version. Make sure that you use an OJDBC6 driver to perform data domain discovery on the Spark engine to improve the performance.

When I run a profile to perform a data domain discovery on the Spark engine, data domain discovery fails with an error message: `org.apache.hadoop.security.authorize.AuthorizationException: User: <infa_user> is not allowed to impersonate <hadoop_user>`**. How do I resolve this issue?**

The issue occurs due to the absence of impersonation user entries in the Hadoop cluster. To resolve this issue, add proxyuser configuration in HDFS service. See https://kb.informatica.com/solution/23/pages/68/561734.aspx to solve the impersonation issue.

When I run a profile on parquet files that contain the Binary data type for the Microsoft Azure Data Lake Store and Amazon S3 resources, the profile run fails. How do I resolve this issue?

Before you run a profile on parquet files that contain the Binary data type, perform the following steps:

1. In Informatica Administrator, click the **Connection** tab in the **Manage** section.
2. Select the **Hadoop Connection** that you use to run the profile.
3. In the **Spark Configuration** tab, click the edit icon.
4. In the **Advanced Properties** field, add the value: `spark.sql.parquet.enableVectorizedReader=false`
5. Click **Save**.
6. Run the profile in Catalog Administrator.

What are the prerequisites for profiling warehouse to support data discovery on Spark runtime environment?

To run profiles on the Spark engine and write the profiling results to the profiling warehouse, you need to configure the JDBC v2 connection for the profiling warehouse.

Before you run profiles on the Spark engine, perform the following steps:

1. Download the JDBC v2 jar file which you used to configure the profiling warehouse JDBC connection. For example, `com.informatica.datadirect.dworacle_5.1.4.G`.
2. Copy the file to the `<INFA_HOME>/externaljdbcjars` location.
   If you used the data direct driver class `com.informatica.jdbc.oracle.OracleDriver` to configure the profiling warehouse, use the keyword **com.informatica.datadirect** to search for the data direct jars in `<INFA_HOME>` directory. Copy the file to the `<INFA_HOME>/externaljdbcjars` location.

To run profiles on the Spark engine, what sources and connectivity type do I use to configure for the profiling warehouse?

The following table lists the databases and connectivity type required for profiling warehouse to support Spark runtime environment for profiles:

Database	Connectivity type	Supported? Certified?
Oracle	JDBC **Note:** You require JDBC v2 license.	Supported, certified. **Note:** The Catalog Service converts the JDBC connection to JDBC v2 connection because JDBC in Oracle does not support data discovery on Spark.
MS SQL Server	JDBC	Supported, not certified
IBM DB2	JDBC	Yes

Why do I need to manually copy the JDBC driver from one directory to another even though the drivers are part of the installer for data discovery on Spark?

Unlike native runtime environment, there are no JDBC drivers in the Spark execution class path. To run profiles on the Spark engine, you need a JDBC V2 connection to the profiling warehouse. To use the JDBC connector, you need to place the jar files in the `externalJdbcjars` location.

What are the prerequisites to run data domain discovery profiles on the Spark engine?

For information about prerequisites to run data domain discovery on the Spark engine, see https://docs.informatica.com/data-catalog/enterprise-data-catalog/10-4-1/_catalog-administrator-guide_enterprise-data-catalog_10-4-1_ditamap/managing_data_domains/data_domain_discovery_on_the_spark_engine.html.

What type of connections can I use for the reference database to run profiles on the Spark engine?

The Content Management Service uses native connection. For more information about staging reference tables on the cluster, see https://docs.informatica.com/data-catalog/enterprise-data-catalog/10-4-1/_catalog-administrator-guide_enterprise-data-catalog_10-4-1_ditamap/managing_data_domains/data_domain_discovery_on_the_spark_engine.html.

What data discovery functions are supported by the Spark engine?

Function	Supported
Single data object column profile	Yes
Single data object data domain discovery	Yes
Scorecard metrics	Yes
Drill down on inferred values and value frequencies	Yes
Export profile results	Yes
Single data object profile with primary key discovery	No
Single data object profile with functional dependency discovery	No
Column profile with *Sample first <number> rows* sampling option	No
Column profile with *Random sample <number> rows* sampling option	No

Are operating system profiles supported for data discovery on the Spark engine?

Yes, operating system profiles are supported for data discovery on the Spark engine.

Is Hive on Amazon S3 or Microsoft Azure Data Lake Store supported on the Spark engine or is there a dependency on another Hadoop distribution?

Hive is supported on Amazon S3 or Microsoft Azure Data Lake Store but not certified. There are no dependencies on any other Hadoop distribution.

Is HTypes supported for data discovery on the Spark engine?

No, HTypes is not supported for data discovery on Spark engine.

Are objects like TDCH, Oraoop, or PWX connectors supported for data discovery on the Spark engine?

No, objects such as TDCH, Oraoop, or PWX connectors are not supported for data discovery on the Spark engine.

Is there a CLI utility or engine switching utility from Blaze to Spark for profiles ?

No, there is no CLI or engine switching utility from Blaze engine to Spark engine. As a workaround, you can run the profiles on the Spark engine.

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